

Surprises, Shapes or Machines?

Forecasting the Market’s Response to Macroeconomic Releases with Dynamic Time Warping (DTW)

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Abstract

Do past market reactions to scheduled macroeconomic releases help forecast the next move? We match the price path around each release to earlier paths with dynamic time warping (DTW) and trade the move that followed the closest matches in four futures markets: the S&P 500 (ES), Nasdaq-100 (NQ) and Dow (YM) equity indices and the 10-year Treasury note (ZN). The dispersion of the matches predicts the size of the next move; their direction is weaker and changed sign around 2018. On 2021–2026 the ES book earns a Sharpe ratio of 0.64 after costs with no market beta, NQ only 0.11, and YM loses (−0.39). The DTW book diversifies surprise-based trading. ZN reacts, but its tick size absorbs the edge.

JEL: G14, G13, C14, E44

Keywords: dynamic time warping; analogue forecasting; macroeconomic announcements; futures; event-driven trading; transaction costs

Introduction

Scheduled macroeconomic releases are among the cleanest information events in financial markets. Their timing is announced months in advance, and futures prices react within the first minute (Fleming and Remolona, 1999; Balduzzi et al., 2001; Andersen et al., 2003). In equity-index futures, however, part of the move continues well after that minute: after a hot CPI print the E-mini S&P 500 (ES) falls about 37 basis points (bps) in the first minute and a further 17 bps over the next half hour. This paper asks whether the *shape* of the market’s reaction to a release, compared with its reactions to earlier releases, forecasts that next move.

We match the price path around each release to past paths with dynamic time warping (DTW) and trade the move that followed the closest matches. The book trades five back-to-back 30-minute windows after every release that moves the market, enters only after the release is public, and is scored on the held-out years 2021–2026 after costs in four futures markets: the S&P 500 (ES), Nasdaq-100 (NQ) and Dow (YM) equity indices and the 10-year Treasury note (ZN).

Three findings stand out. First, the dispersion of the matches forecasts the size of the next move in every market and year, while their direction is weak and changed sign around 2018. Second, the ES book earns a Sharpe ratio of 0.64 at a quarter tick per side with no market beta, but NQ earns only 0.11 and YM, added later as an out-of-sample market, loses (−0.39). Third, the edge is concentrated in

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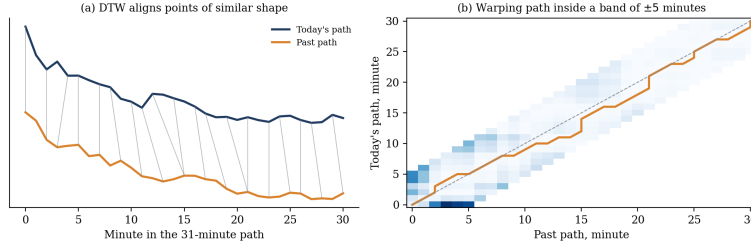


Figure 1. How DTW compares two price paths (stylized example). Panel (a): the lines connect the points that DTW aligns; the past path reaches the same low about two minutes later. Panel (b): the local cost inside the band $|i - j| \leq 5$ and the optimal warping path (orange); the dashed diagonal is the minute-by-minute (Euclidean) alignment.

the first window after the release, and in ZN it is smaller than one tick. The Sensibility part explains the approach, the Data and Model parts describe the inputs and the strategy, and the Backtest part reports the results.

Sensibility

Markets rarely react to the same news on the same clock. The shape of the repricing repeats, but the timing may differ by a few minutes from one release to the next. A distance metric that cannot see through that slippage will call two identical reactions different. Dynamic time warping (DTW) measures the similarity between two time series while letting their time axes stretch and compress, so two paths are close if they trace the same shape, even when one runs faster or lags the other (Sakoe and Chiba, 1978; Berndt and Clifford, 1994). This feature makes DTW well suited to price data: Euclidean distance compares minute by minute, and correlation is just as rigid in time. We use DTW as an analogue forecaster (Lorenz, 1969). We rank historical price paths by their similarity to a target window and read off the returns that followed. We posit that similar price paths have similar short-term forward returns. Figure 1 illustrates the alignment.

Applied naively, DTW runs into two problems. The first is cost: comparing a target against every window in years of minute-level data is slow. The second is that, with a large enough library, some historical paths will match the target by pure luck, and a lucky match says nothing about why the path formed. We address both issues by restricting the library to windows that follow scheduled macroeconomic releases such as CPI and the Jobs Report, events macro traders reliably react to. The library shrinks, and every match shares a common root: one public shock at a known time, with participants repricing rate expectations together. Our refined hypothesis is that similar price paths following scheduled macro releases have similar short-term forward returns. For example, take the price path following a CPI release. Compared against past paths following other releases, such as the Jobs Report, retail sales or PCE, most will strongly differ, yet a few will trace nearly the same shape. Those few are far more likely to reflect the same repricing than sheer luck.

Data

I. Bloomberg macroeconomic events

The event calendar is the Bloomberg U.S. economic release calendar from 2010 to 2026. It contains 179 release lines—a line is one published statistic, such as “CPI MoM” or “Change in Nonfarm Payrolls”—and 27,137 release rows, each with the release date and time in Eastern Time, the actual value and the median of the Bloomberg survey of forecasters. Most price and labor-market data are released at 08:30, survey indicators such as the ISM indices at 10:00, and FOMC decisions at 14:00.

Table 1. Contracts.

	ES	NQ	YM	ZN	ZT
Underlying	S&P 500	Nasdaq-100	Dow Jones Ind.	10-year note	2-year note
Continuous series	calendar roll	calendar roll	calendar roll	volume roll	—
Tick size	0.25 points	0.25 points	1 point	1/64 point	—
Value of one point	\$50	\$20	\$5	\$1,000	—
Value of one tick	\$12.50	\$5.00	\$5.00	\$15.625	—
One tick, in bps of contract value	≈ 0.5	$\approx 0.1\text{--}0.3$	$\approx 0.25\text{--}1$	≈ 1.4	—
Role in this paper	traded	traded	traded	cost-gated	dropped

Notes: Databento GLBX.MDP3, one-minute bars, July 2010 to June 2026. ZT was pulled but dropped: before 2019, fewer than 7% of its release windows have a trade in every minute, and a check against tick-level trades confirmed that the gaps are minutes without trading, not missing data.

The DTW model uses the calendar for its timing, not its values: a release is an instant T at which information arrives, and the model compares the market’s reaction at T with its reactions at earlier releases. The actual values and the consensus are not used.

Many lines are released together as parts of one report. The CPI report alone produces eight lines at the same second, and the Employment Situation report twelve. Trading each line separately would trade the same minute several times, so Model, Section I groups the lines into release families before any other step. In the family construction, 163 lines collapse into 67 families.

II. Futures data (S&P 500, Nasdaq-100, Dow, 10-year and 2-year notes)

We use one-minute OHLCV bars from Databento (CME Globex, GLBX.MDP3) from July 2010 to June 2026. Table 1 summarizes the contracts. ES, NQ and YM are continuous front-month series rolled on the calendar; ZN is rolled on volume. For ES, the first years of the vendor file had many incomplete sessions, and we re-pulled 2010–2015 so that the whole sample has complete sessions. YM (\$5 per index point, tick one point) is the only other U.S. equity-index future on the same feed with one-minute history from 2010. Its file has 5.2 million one-minute bars; cleaning keeps 3,754 sessions (84–95% of each year’s sessions), and 4.8% of the kept minutes are filled with the last price, 8–15% before 2016, like early NQ, and under 2% from 2018.

Sessions. A trading session runs from 18:00 to 17:00 Eastern Time and is dated by the day on which it closes. We drop exchange holidays and keep a session only if it has at least 85% of the asset’s median number of bars per weekday session in that year; the threshold is relative because ZN and early NQ and YM trade fewer minutes than ES. Minutes without a trade inside a kept session are filled with the last price. Bars are labelled by their start: the bar labelled 08:30 covers 08:30:00–08:30:59.

Rolls. A path or trade that straddles a change of contract would include the calendar-spread jump. We drop any game in which the contract changes between the first minute of its path and its exit, and buy-and-hold benchmarks skip the price jump at each roll.

Figure 2 shows the equity and Treasury futures over the sample, including YM.

A. Why we chose to look at these

All five contracts are among the most liquid futures in the world, trade nearly around the clock on CME Globex, and react to the same U.S. releases, so one calendar and one session definition serve them all. They differ in how they should react. A hot inflation print raises the expected path of the policy rate and lowers the present value of future earnings, more so for the long-duration technology stocks that dominate NQ; the Dow behind YM holds fewer technology and more industrial and financial

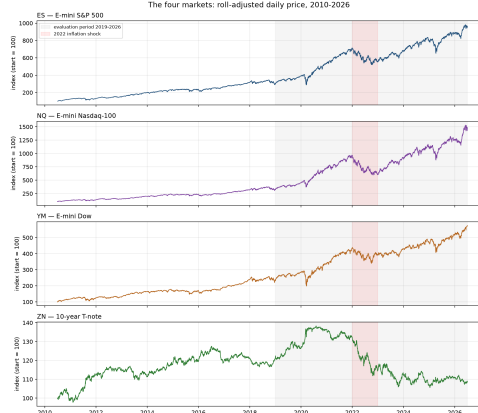


Figure 2. ES, NQ, YM and ZN, 2010–2026. Roll-adjusted daily prices rebased to 100. Grey: the 2019–2026 period; red: the 2022 inflation shock.

stocks than the S&P 500, so it tests whether what works in ES carries to a different mix of the same market. The Treasury notes respond to the same news through yields: the CPI surprise explains about 30% of the five-minute reaction of ZN, as it does for ES. What separates the markets for a strategy that trades often is the tick: one tick is about 0.5 bps of contract value in ES, 0.1–0.3 bps in NQ and 0.25–1 bp in YM, but about 1.4 bps in ZN, large next to its 30-minute moves. ES and NQ are the core of the book; YM was added after their results were known, as an out-of-sample market; ZN is a candidate that a cost gate admits or excludes (Model, Section III.C).

B. Why we dropped the 2-year

The 2-year note future (ZT) is the most direct read on the expected policy rate and would be the natural market for rate news. But a path-based method needs a price in every minute of the path, and before 2019 fewer than 7% of ZT’s release windows have a trade in every minute. A fresh pull compared against tick-level trades matched exactly, so the gaps are minutes in which ZT did not trade, not missing data. With a library that sparse before 2019, the analogues cannot be built, so we dropped ZT at the data stage.

III. Data for DTW: the event panel

The DTW model does not use the bars directly. It uses a panel of *games* built from them, one row for each asset, release family, release time and window (Model, Section I, Model, Section III.B). Each row carries three objects, all taken from the cleaned one-minute closes of Data, Section II:

- **The path:** the 31 closes ending at the entry minute, standardized within the window (equation (2)). For the first window the path ends at $T + 1$, so it holds the last half hour before the release and the first minute of the reaction.
- **The target:** the log return from the close of the entry minute to the close 30 minutes later, the return that the trade earns.
- **The labels:** the family, the clock slot, the window, the contract and whether the family passes the impact screen in that year (the *admitted* flag).

A game is kept only if every minute of its path and of its holding window is inside a kept session, the release minute and the minute after it have a price, and the same contract is traded from the first minute of the path to the exit. A missing minute is never interpolated; the game is dropped.

Table 2. The DTW event panel, 2010–2026.

	ES	NQ	YM	ZN
Clean one-minute bars (millions)	5.39	5.26	5.20	5.39
Games, all 67 families	57,146	55,317	54,455	56,665
Families admitted in some year	8	8	5	12
Admitted games with DTW features	3,420	2,735	2,390	5,420
of which scored in training, 2013–2017	820	525	640	1,085
of which scored in validation, 2018–2020	840	665	460	1,130
of which in the test years, 2021–2026	1,760	1,545	1,290	3,090
Trades in the test years, 2021–2026 (after the gate)	1,202	1,141	870	1,945
Median analogue dispersion σ_e (bps)	17.8	24.1	17.0	6.1

Notes: A game is one (asset, family, release, window). All games form the library of candidate analogues; admitted games are those whose family passes the impact screen in that year, and only they are traded. Trades in the test years are admitted games that pass the gate of equation (5), after games at the same minute are merged.

The release times come from the Bloomberg calendar (Data, Section I), after the 163 release lines are grouped into 67 families.

Every game, admitted or not, enters the library of candidate analogues; only admitted games are traded. The library is therefore much larger than the traded set: on ES there are about 57,000 games in 2010–2026 but 3,420 admitted ones (Table 2). The first three years, 2010–2012, are a burn-in period: they fill the library and give the impact screen its 30 earlier releases, but no game in them is scored. The DTW features are computed for admitted games only, from strictly earlier games in the library, so the panel can be built once and then split by year into the training, validation and test samples of Backtest, Section I. The panel for the four assets has 223,583 rows.

The panel is small where it matters. Training scores 820 ES games and validation 840, and the signal we look for is a correlation of a few hundredths; this is why the setting is fixed in advance and checked once (Backtest, Section II). The panel is also uneven across assets: the impact screen admits more families on ZN, because the Treasury market reacts to more of the calendar, and ZN’s analogues disagree by a third as much as ES’s in bps, in line with its smaller moves. YM has the fewest admitted families: CPI, payrolls, the ISM Manufacturing and Services indices and the FOMC Minutes, the core of the ES list without Core PCE, Retail Sales and Wholesale Inventories.

Model

The model decides which releases to trade (Model, Section I), turns the price path at entry into a forecast (Model, Section II) and turns the forecast into positions (Model, Section III). Nothing in it uses data at or after the entry minute.

I. Event Universe Construction

Release lines that share at least 80% of their timestamps are grouped into one *family*, so each report (CPI, the Employment Situation) is one event. A family is admitted for asset a in year y if it moves the market more than other releases at the same clock slot. With $m_T = |p_{T+30} - p_T|$ the absolute 30-minute move after minute T ,

$$I_{f,a,y} = \overline{m}_{T \in \mathcal{R}_f} / \overline{m}_{T \in \mathcal{C}_f}, \quad (1)$$

where $\mathcal{R}_f$ are the family’s earlier releases and $\mathcal{C}_f$ the same clock slot on days when another family releases there but f does not. A family needs 30 earlier releases and a Welch t -statistic of at least 2 on the log moves; the screen is redone each year from earlier years only. On ES it admits eight families (payrolls, CPI, ISM Manufacturing and Services, FOMC Minutes, Retail Sales, Core PCE, Wholesale

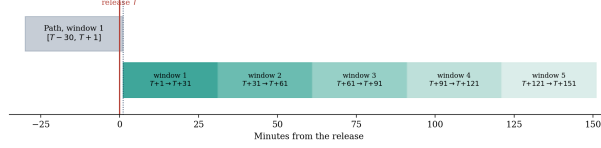


Figure 3. The anchor ladder: five 30-minute windows after a release at T . Each window's path is the 31 minutes ending at its entry.

Inventories). The FOMC decision fails, because at 14:00 its only control is the Minutes. Each admitted release is traded in five windows; one (asset, release, window) is a *game*.

II. Signal

For game e with entry minute τ_e , the path is the 31 log prices ending at entry, standardized,

$$q_{e,i} = (p_{\tau_e-31+i} - \bar{p}_e)/s_e, \quad i = 1, \dots, 31, \quad (2)$$

so DTW compares the shape of the reaction, not its size; the target is the next 30-minute return y_e . Candidates are earlier games on the same asset, clock slot and window, from any family, that exited before e entered, within three years and at most the 600 most recent (at least 20 are required). The DTW distance d_j to each candidate uses squared local costs and a Sakoe–Chiba band of five minutes (Keogh and Ratanamahatana, 2005). The $K = 15$ nearest neighbours $\mathcal{K}_e$ are weighted by similarity and age α_j (days), $\omega_j \propto d_j^{-1} e^{-\alpha_j/730}$, and give two features:

$$\mu_e = \sum_{j \in \mathcal{K}_e} \omega_j y_j, \quad \sigma_e = \left(\sum_{j \in \mathcal{K}_e} \omega_j (y_j - \mu_e)^2 \right)^{1/2}. \quad (3)$$

The mean μ_e forecasts the direction of the next move; the dispersion σ_e measures how much the analogues disagreed, and forecasts the move's size (Backtest, Section II). The setting was chosen on ES in an earlier version of the model and is used unchanged for the other assets.

III. Model

A. How do we turn this signal into a strategy

The position in game e is a direction, a gate and a size, $w_e = \text{sign}(\mu_e) G_e S_e$, held for 30 minutes from the close of the entry minute. Games of different families at the same minute are merged into one position. With n ticks of slippage per side, tick δ and entry price P_e , the net return in bps is

$$r_e = w_e y_e - |w_e| \cdot 2n \delta / P_e \times 10^4, \quad n \in \{0, \frac{1}{4}, \frac{1}{2}\}. \quad (4)$$

B. Why 5 windows?

Each release is traded in five back-to-back 30-minute windows entering at $T+a_k$, $a_k \in \{1, 31, 61, 91, 121\}$ (Figure 3). The reaction to a large release lasts more than 30 minutes; one window gives too few trades to measure a small edge (five give about 225 a year in ES); and the windows test where the information is: if the path carries news, the first window should be best (Backtest, Section III). No window enters at T , whose price predates the release.

Table 3. Sample split.

Period	Years	Use
Burn-in	2010–2012	library of past reactions and screen history only
Train	2013–2017	choosing the DTW setting, gate and sizing
Validation	2018–2020	checking that the choice holds
Test	2021–2026	results

C. Gating

The forecast is noisy near zero, so the book trades only its tails,

$$G_e = \mathbf{1}\{\mu_e \leq Q_{0.3} \text{ or } \mu_e \geq Q_{0.7}\}, \quad (5)$$

with quantiles fixed on the asset’s 2013–2020 games: the outer 60% of forecasts are traded.

D. Weighting

Trades are sized inversely to the dispersion, $S_e = \min(\tilde{\sigma}/\sigma_e, 2)$, with $\tilde{\sigma}$ its median in the fitting years: consistent analogues trade up to two contracts, scattered ones less than one. Assets are combined with inverse-volatility weights reset monthly,

$$R_t = \sum_a \lambda_{a,t} R_{a,t}, \quad \lambda_{a,t} \propto G_{a,t}^c / \hat{v}_{a,t}, \quad (6)$$

where $\hat{v}_{a,t}$ is the 60-session volatility of the futures and $G_{a,t}^c$ a cost gate: an asset is traded only while a round trip costs less than 10% of its median 30-minute move on admitted releases over the past year. ES and NQ always pass (weights about 58% and 42%), as does YM (about 36/26/38% with ES and NQ); ZN passes only in part of 2023 (Backtest, Section V).

Backtest

I. What did we test

We trade the model of the Model part, with its setting and rule fixed in advance, and score it on the held-out years. Returns are in bps of one contract’s notional, summed rather than compounded, on a daily grid of trading sessions with zero on days without a trade; return, volatility and the Sharpe ratio are annualized over 252 sessions. Costs are charged per side at a quarter or a half tick.

A. Time period

Train, validation and test split. Table 3 shows how the years are used. The first three years only fill the library of past reactions and the history the universe screen needs. Settings are chosen on the training years, checked on the validation years, and the test years are used to report results. The gate cut-offs and the size reference come from 2013–2020, so nothing in the test years is fitted.

Why 30 minutes. Each trade is held for 30 minutes and each path is the 31 minutes ending at the entry. The horizon matches how long the news keeps moving prices: after hot CPI prints ES falls about 37 bps in the first minute and a further 17 bps by $T+30$, and is roughly flat over the following half hour; NQ and YM behave the same way. A 30-minute path is long enough to contain the whole first reaction and short enough that the library of past releases at the same clock time holds hundreds of comparable

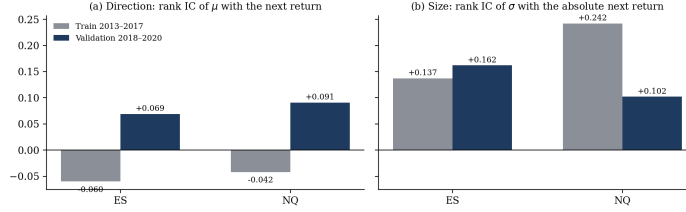


Figure 4. Information coefficients of the DTW features on admitted games, ES and NQ. Panel (a): rank correlation of μ with the next 30-minute return. Panel (b): rank correlation of σ with the absolute next 30-minute return.

paths. Equal path and holding lengths also make the five windows of the ladder back-to-back, so no minute is traded twice.

B. Assets

Each asset is first traded on its own with the same setting, gate and sizing: ES and NQ, the core of the book; YM, run through the identical pipeline after the ES and NQ results were known, as an out-of-sample market; and ZN. Two multi-asset books follow.

C. Equities book

The equities book combines ES and NQ with the inverse-volatility weights and cost gate of equation (6); both always pass the gate at a quarter tick, and the weights average 58% ES and 42% NQ. We also report the three-index book ES + NQ + YM, with weights of about 36%, 26% and 38%.

D. Combined book

The combined book adds ZN on the same terms, and the cost gate decides whether it is traded. Before costs every asset passes and ZN takes about 60% of the weight; at a quarter tick a round trip is 10–17% of ZN’s typical 30-minute move, so it is excluded except in part of 2023 (Backtest, Section V).

II. General Statistics

Signal quality. Before trading, we ask whether the two features rank what happened next (Figure 4). The direction forecast μ ranks the next return *negatively* in the training years (−0.060 in ES, −0.042 in NQ) and *positively* in the validation years (+0.069 and +0.091); in YM and ZN it is close to zero in both (−0.006 and +0.018; −0.003 and −0.005). The dispersion σ , in contrast, ranks the absolute size of the next move positively in every market and period, at 0.10–0.24. The analogues know how large the next move will be; what they say about its direction is small and changed sign around 2018.

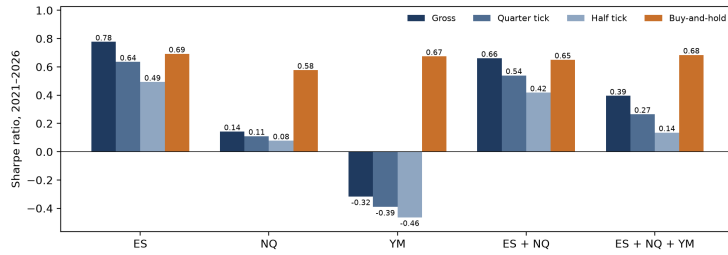
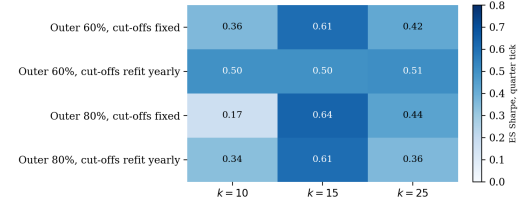
Results. Table 4 reports the test years. The ES book trades 1,202 times, about 225 a year, and earns a Sharpe ratio of 0.78 before costs, 0.64 at a quarter tick and 0.49 at a half tick, against 0.69 for holding ES, with a drawdown of −5.0% against −31.1% and a beta of 0.007. Half its trades win and 53% are long, so it is not a disguised long position. NQ is positive but weak (0.11 at a quarter tick). YM does not work: only five families pass its screen, and it loses before costs (−0.32) and after them (−0.39). The equities book earns 0.54 with a drawdown of −5.3%; adding YM lowers it to 0.27. The combined book earns 0.40 (Figure 5a).

Checks. Table 5 reports five checks. Rebuilding the features with the path running five minutes *past* the entry raises the Sharpe ratio to about 2.8 in every market, so the real book’s timing is sound. Drawing the direction of each trade at random beats the ES book only 4.1% of the time, but beats

Table 4. General statistics of the DTW event book, test years 2021–2026, quarter tick per side.

Book	Trades	Sharpe ratio			Buy-and-hold	Total return	Ann. vol	Max DD	Hit rate
		Gross	$\frac{1}{4}$ tick	$\frac{1}{2}$ tick					
ES	1,202	0.78	0.64	0.49	0.69	10.9%	3.2%	−5.0%	50.0%
NQ	1,141	0.14	0.11	0.08	0.58	2.1%	3.6%	−8.2%	49.5%
YM	870	−0.32	−0.39	−0.47	0.67	−4.8%	2.4%	−8.0%	50.0%
ZN	1,945	0.34	−0.97	−2.24	−0.69	−7.4%	1.4%	−8.0%	46.9%
Equities (ES + NQ)	2,343	0.66	0.54	0.42	0.65	7.4%	2.6%	−5.3%	49.8%
ES + NQ + YM	3,213	0.39	0.27	0.14	0.68	2.8%	2.0%	−3.0%	49.8%
Combined (ES + NQ + ZN)	2,476	0.76	0.40	0.41	0.24	5.4%	2.5%	−5.3%	49.3%

Notes: Direction $\text{sign}(\mu)$, outer 60% of μ with cut-offs from 2013–2020, size $\min(\tilde{\sigma}/\sigma, 2)$, five windows per admitted release. Return, volatility, drawdown and hit rate at a quarter tick. Multi-asset books use the weights and cost gate of equation (6), so each cost column is the book a trader facing that cost would run. Buy-and-hold drawdowns: ES −31.1%, NQ −48.3%, YM −25.6%.

**(a)** Sharpe ratio by cost level, against buy-and-hold.**(b)** ES Sharpe ratio at a quarter tick by K , gate width and cut-off rule.**Figure 5.** The DTW event book, 2021–2026. Panel (b): cut-offs are either fixed from 2013–2020 or recomputed each year.**Table 5.** Checks on the DTW event book, 2021–2026, quarter tick per side.

	ES	NQ	YM	ES + NQ	ES + NQ + YM
Leak test: Sharpe with the path 5 minutes past entry	2.86	2.85	2.84	3.62	3.83
Random-sign test: p -value, 5,000 draws	0.041	0.37	0.76	0.064	0.18
Always long in the same windows: total return	−2.4%	−1.0%	−6.2%	−1.8%	−3.6%
Break-even cost, ticks per side	1.36	1.10	—	1.35	0.76
Rules with Sharpe > 0, of 12	12	7	1	12	12
Appraisal ratio vs buy-and-hold	0.61	0.11	−0.38	0.53	0.26

Notes: YM loses before costs, so it has no break-even cost. The YM columns come from the same code run with YM added, in which the ES, NQ and equities results reproduce exactly.

NQ and YM often ($p = 0.37$ and 0.76). Holding the same windows always long loses in every book, so the ES profit comes from direction. ES breaks even at 1.36 ticks per side. Across twelve gate, cut-off and sizing rules ES and the equities book are always positive, YM only once, and across the number of neighbours K (Figure 5b) the ES Sharpe ratio ranges from 0.17 to 0.64. Broader searches over about 138,000 settings confirm that the direction is unstable across periods (the Sharpe ratios of the same settings on 2014–2017 and 2018–2020 correlate at 0.15). The appraisal ratio—alpha on buy-and-hold over residual volatility—is 0.61 for ES and 0.53 for the equities book; with a directional IC of 0.027 on about 225 bets a year, the fundamental law of active management (Grinold and Kahn, 2000) implies about 0.4, so 0.4–0.6 is a fair range.

III. Hit rate per window

Table 6 and Figure 6 split the book by window. The first window, entering at $T + 1$, is the best in every equity index: it wins 53% of its trades in ES, 53% in NQ and 58% in YM, with Sharpe ratios of

Table 6. Hit rate and Sharpe ratio by window, 2021–2026, quarter tick per side.

Entry	$T + 1$	$T + 31$	$T + 61$	$T + 91$	$T + 121$
ES hit rate (Sharpe)	53.4% (0.95)	50.2% (0.27)	52.1% (0.33)	50.0% (0.45)	43.4% (−0.91)
NQ hit rate (Sharpe)	53.3% (0.38)	49.3% (0.02)	50.0% (−0.18)	48.5% (−0.04)	45.8% (0.01)
YM hit rate (Sharpe)	58.1% (0.41)	47.8% (−0.40)	46.2% (−0.17)	52.1% (−0.18)	44.7% (−0.57)
ZN hit rate (Sharpe)	47.3% (−0.39)	48.5% (−0.41)	46.9% (0.02)	45.8% (−0.64)	45.7% (−0.94)

Notes: Trades per window: 150–262 in the equity indices and 339–442 in ZN.

0.95, 0.38 and 0.41, against 1.7%, 5.4% and −3.0% for always long in the same windows. The hit rate then falls toward a coin flip in the second to fourth windows, and the fifth, at $T + 121$, is the worst everywhere: 43% in ES (Sharpe −0.91), 46% in NQ and 45% in YM. In YM the first window is the only one that makes money, which is why its five-window book loses; in ZN no window reaches 50%. The path carries information while it still contains the reaction to the release, and less once the reaction is over. Choosing the window on these test years would be choosing on the test period, so the book keeps all five.

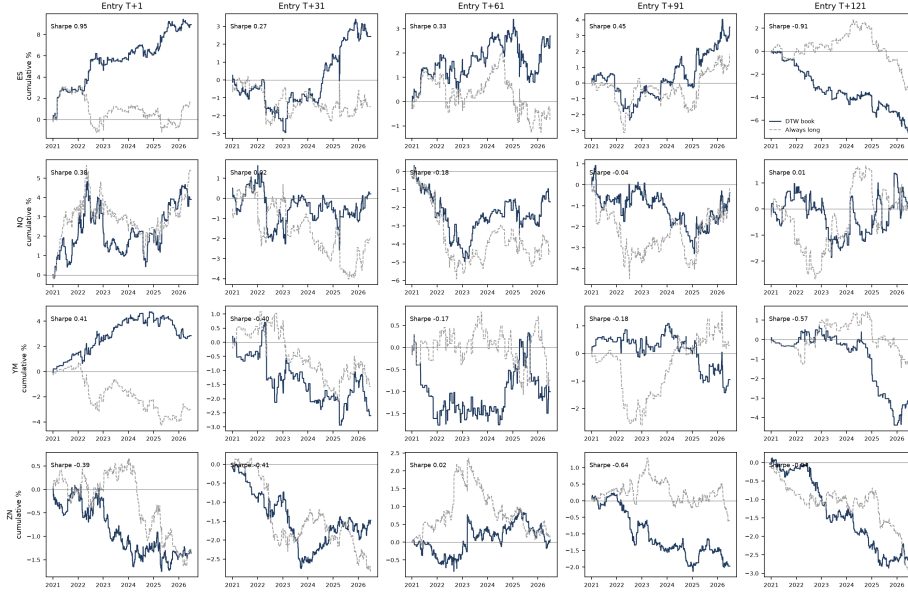


Figure 6. Cumulative net P&L at a quarter tick by window, 2021–2026, against always long in the same windows. Rows: ES, NQ, YM, ZN; columns: entry at $T + 1$, $T + 31$, $T + 61$, $T + 91$ and $T + 121$.

IV. 2022 for equities

Table 7 splits the test period by year. The ES and equities books are positive in five of six years; 2022, the year of the inflation shock and the fastest monetary tightening in four decades, is the exception (equities −0.87, −2.5%). A month-by-month breakdown shows that 2022 was not a period of poor matches: the mean weighted similarity of the neighbours was about 0.51 in ES and 0.50 in NQ in the first half of 2022, the same as in other years. In those months the release windows themselves fell (about −10 bps on average in ES and −12 bps in NQ), the ES book was long in about 63% of its trades, and the worst month, May, was a short position into a rebound. The paths matched as closely as usual; what those paths had done next in earlier years no longer held, because the market’s reading of the same data had changed with the Federal Reserve’s reaction function. YM moved the other way: it gained in 2022 (0.34) and lost most in 2025. Win rates are 47–54% in every year, so the edge, where it exists, comes from winners being larger rather than more frequent.

Table 7. Sharpe ratio by year, quarter tick per side.

	2021	2022	2023	2024	2025	2026
ES	0.67	-0.48	0.89	1.33	0.76	1.27
NQ	0.12	-0.84	-0.06	-0.28	0.94	1.36
YM	-0.23	0.34	0.44	-0.28	-1.67	-0.61
ZN	-1.24	-1.01	-1.73	-0.52	-0.25	-0.89
Equities (ES + NQ)	0.71	-0.87	0.70	0.78	1.04	1.58
return	+1.3%	-2.5%	+1.6%	+1.7%	+3.1%	+2.3%
ES + NQ + YM	0.46	-0.47	0.80	0.49	-0.01	1.16
Combined (ES + NQ + ZN)	0.71	-0.87	-0.23	0.77	1.03	1.57

Notes: 2026 is a partial year (to June).

V. Regime change for the 10-year

ZN went through a larger change of regime than the equity indices. Before 2021 policy rates were near zero, ZN’s moves after releases were small, and a round trip at a quarter tick cost 13–21% of its typical 30-minute move in every year. The 2022 tightening made rate news the market’s main driver and ZN’s post-release moves larger: the cost ratio fell from 17% in 2021 to 15% in 2022 and 10% in 2023, before rising again to 12–17% in 2024–2026. Before costs the book does find something in ZN: a gross Sharpe ratio of 0.34 and a small positive directional IC (+0.016). But the regime change did not make the direction tradeable. In 2023, when the larger moves let ZN through the cost gate for part of the year, it lost, and the combined book scored -0.23 that year against 0.70 for equities. ZN loses in every test year at a quarter tick, its strategy tracks being always long in the same windows (-7.4% for both), and it breaks even at only 0.07 ticks per side. Without a gate, fixed equal-risk weights would have given ZN 60% of the combined book at every cost and a Sharpe ratio of -0.23; the cost gate is what keeps the combined book at 0.40. Bigger rate moves did not mean that the past paths forecast the next one: in ZN the tick is larger than the edge in any regime we observe.

VI. Against the surprise strategy

The natural competitor to the DTW book is a strategy that takes its direction from the consensus surprise: buy after a release that beats the median forecast in the direction the market has rewarded, sell after one that misses. Our companion study runs such books on the same four markets, with the same releases, an entry one minute after the release and a 30-minute hold, over 2019–2026 and at a harsher cost of two ticks round trip plus \$4.50 in fees. It also runs a simple DTW-only book at the first window and two books that trade only when the surprise and the DTW forecast agree. These are not the ladder book of this paper—they use one window, a longer sample and a different cost—but they show where the direction of the trade is best taken from.

Figure 7 gives the answer market by market. In all three equity indices the broad surprise book earns about the same net Sharpe ratio, 0.65 in ES, 0.71 in NQ and 0.68 in YM, and the CPI surprise model earns 0.45–0.76. The DTW-only book is weaker in each: it loses in ES (-0.39), is flat in NQ (0.08) and earns 0.47 in YM. DTW adds most as a filter on the surprise. Keeping only the trades on which the two agree raises the Sharpe ratio of the broad book in YM to 0.94 (0.94 also without 2022), and keeping only the CPI trades on which they agree raises ES’s CPI book from 0.76 to 0.93, on fewer trades. In ES the broad agree book is lower than the surprise book alone (0.31), so the filter does not help everywhere. In ZN every book loses after costs, as in Backtest, Section V: the broad surprise book goes from -0.30 before costs to -2.42 after them.

Figure 8 combines the surprise books across markets. The three equity books rise together, mostly in 2022, and the ES + NQ + YM portfolio earns a Sharpe ratio of 0.78 (37.5% in total) against 0.72 for ES + NQ. ZN loses steadily, and because equal-risk weights give it the largest weight, it turns the

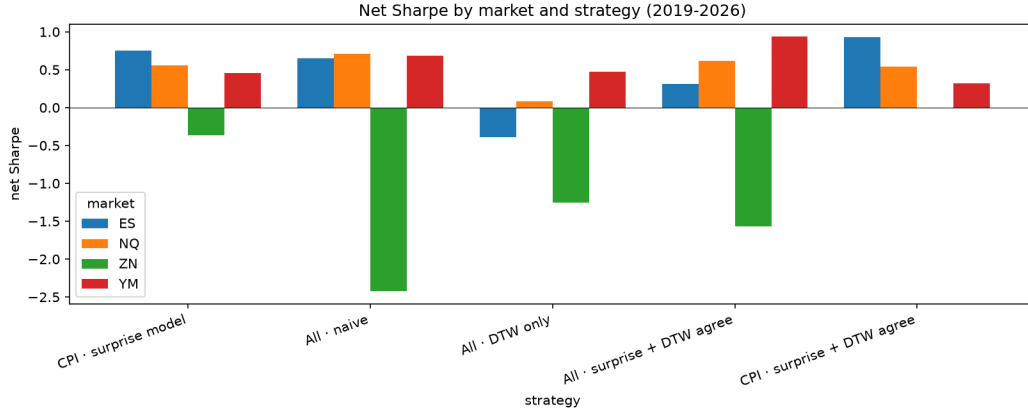


Figure 7. Net Sharpe ratio by market and strategy, 2019–2026, two ticks round trip plus \$4.50. “All”: every release family that passes the impact screen; “CPI”: CPI releases only; “naive”: trade the sign of the surprise; “surprise model”: trade a fitted drift forecast; “DTW only”: trade the sign of the DTW forecast at the first window; “surprise + DTW agree”: trade the surprise only when the DTW forecast has the same sign. ZN has no CPI agree trades.

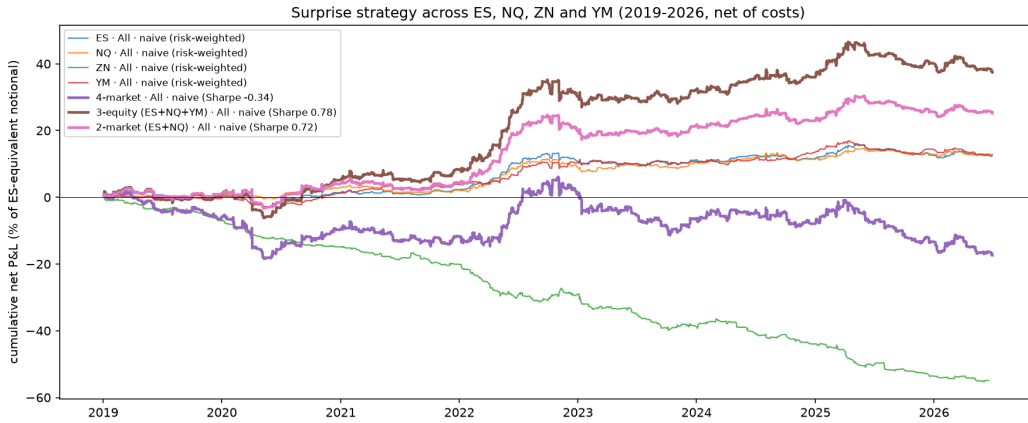


Figure 8. Cumulative net P&L of the broad surprise book (“All · naive”) by market, risk-weighted, and of the two-, three- and four-market portfolios with equal-risk weights fixed on 2016–2018 (ES 1.00, NQ 0.71, YM 1.02, ZN 2.67), 2019–2026, net of costs.

four-market portfolio negative (-0.34). Both results match the DTW book: the equity indices carry the edge, adding YM broadens the book, and ZN should be kept out at these costs. The comparison also locates the DTW book’s role. As a source of direction it is weaker than the surprise in every market; as a second opinion on the surprise and as a book whose returns are nearly uncorrelated with it, it is useful.

Figure 9 shows how these books earn over time, on all screened releases and on CPI alone, and adds a fourth book that blends the two signals instead of requiring agreement. In ES and NQ the surprise-only book is the highest curve on the broad set of releases, and the DTW-only book drifts down in ES from 2022 and goes flat in NQ after 2023, the same break as in Backtest, Section IV. YM is the one market where DTW adds to the surprise: the blend earns a Sharpe ratio of 1.06 and the agree book 0.94, against 0.68 for the surprise alone, with a smoother path and without the 2020 drawdown of the surprise book. On CPI the four books are close in every equity index and most of their P&L comes from 2022, so the CPI comparison rests on a single year. In ZN all four curves fall steadily; the surprise book falls most because it trades every release and pays the cost each time.

The surprise books above are built on the broad set of screened releases. Our first walk-forward study instead trades only the three largest reports—CPI, Core PCE and payrolls—with five rules: the naive sign of the surprise, a fitted surprise model, and three versions of that model conditioned on

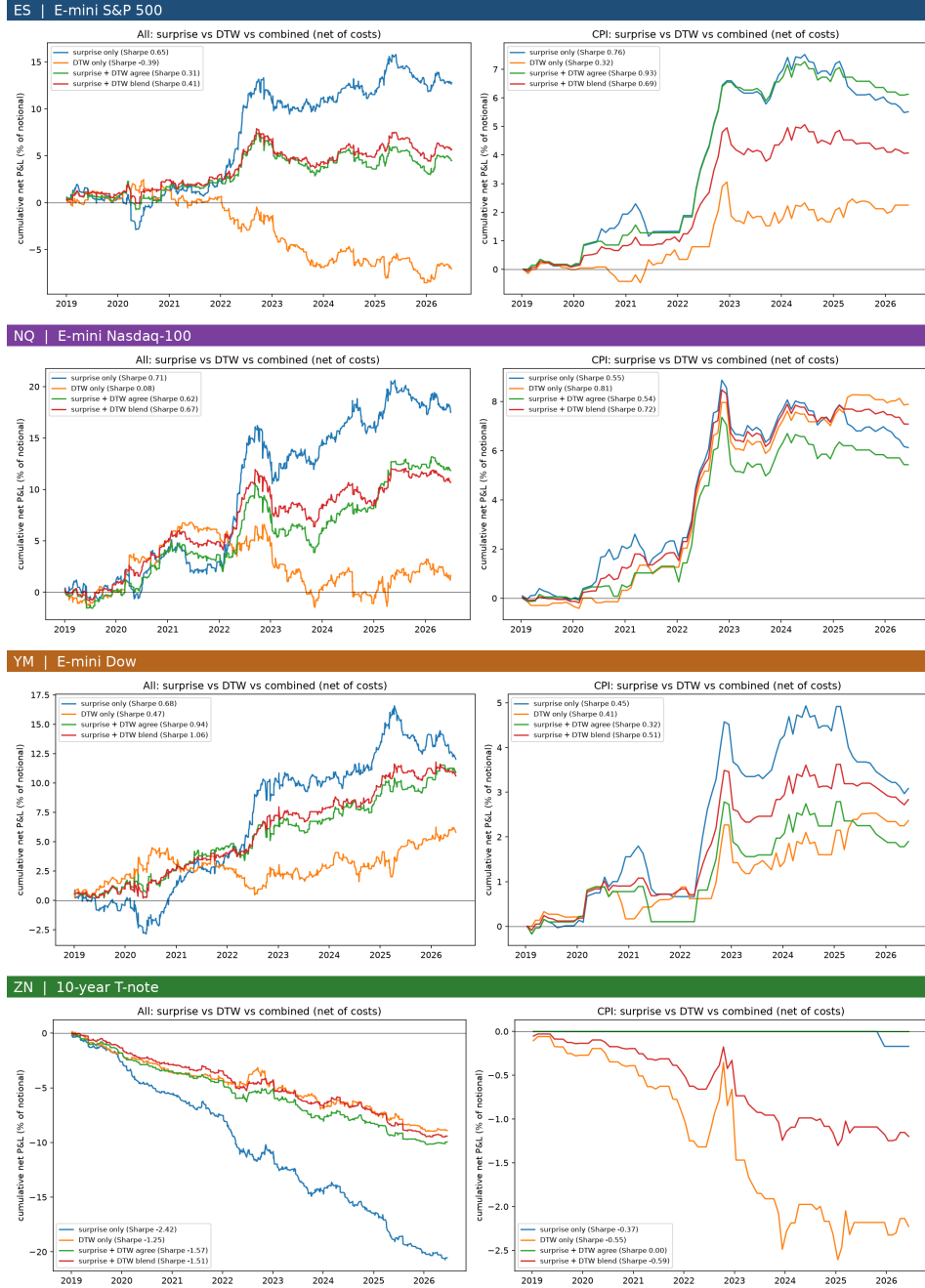


Figure 9. Cumulative net P&L, 2019–2026, two ticks round trip plus \$4.50, of four single-window books: surprise only, DTW only, surprise + DTW agree, and a blend of the surprise and DTW signals. Left: all screened releases; right: CPI only. Rows: ES, NQ, YM and ZN. ZN has no CPI agree trades (flat green line).

the market’s state before the release (volatility, momentum and the like). Every slope, sign and state model is re-estimated on earlier releases only, and trades pay two ticks plus \$4.50. On CPI alone the naive sign is the best or close to the best in each equity index, with Sharpe ratios of 1.03 in ES, 1.00 in NQ and 0.99 in YM, and conditioning on the state does not improve on it by a significant margin. As in the DTW book, almost all of the gain comes in 2022. Adding PCE and payrolls lowers the naive rule to 0.33 in ES and 0.38 in NQ (0.57 in YM). In ZN the fitted models barely trade, because their forecast rarely exceeds a tick, and the naive rule loses (-0.05 on CPI, -0.65 on the three reports). The surprise, in other words, gives a better direction than the DTW analogues in the equity indices whether it is traded on one report or many, and neither source survives the ZN tick.

Conclusion

We asked whether the shape of the market’s reaction to a scheduled macroeconomic release, matched to earlier reactions with dynamic time warping, forecasts the next move in futures. Restricting the library to post-release windows makes the method fast and gives every match a common cause, and the answer has two parts.

The analogues forecast the size of the next move well. The dispersion of the neighbours’ outcomes ranks the absolute next return in every market and period, and sizing trades inversely to it improves the ES and equities books under every rule we tested.

The analogues forecast the direction of the next move only weakly, and not stably. Their directional information coefficient was negative in 2013–2017 and positive afterwards. With the setting and rule fixed in advance, the ES book earns a Sharpe ratio of 0.64 at a quarter tick on 2021–2026, with no market beta, a drawdown of -5% , a direction that beats random signs and a timing that passes a leak test; the equities book earns 0.54. But NQ earns only 0.11, YM—run through the same pipeline as an out-of-sample market—loses, and the ES result ranges from 0.17 to 0.64 across reasonable constructions. The edge sits in the first window after the release, the 2022 loss came from a change in what similar paths did next rather than from poor matches, and in ZN the larger moves of the new rate regime still left the edge below one tick.

Three conclusions follow for practice. First, anchoring to scheduled events and entering only after the release are what make a path-based method testable at all. Second, DTW is best used for what it measures robustly—the size and uncertainty of the next move—while the direction comes from a source with a clearer economic link, such as the consensus surprise, with which the DTW book is nearly uncorrelated and which it improves as a filter in YM and in ES’s CPI trades (Backtest, Section VI). Third, before any capital is committed, the book needs recorded fills in the minutes after releases, because slippage decides whether a per-trade edge of this size survives. Natural extensions are a book that trades only the first window, averaging the forecasts of the best settings, and a direction signal that combines the surprise with the shape of the reaction.

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